EcoSort: Smart AI Waste Classifier

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1. Project Idea & Problem Statement

Sorting garbage manually takes a long time, costs money, and can be dangerous with items like broken glass or batteries. **EcoSort** is a smart solution built with AI computer vision. It automatically classifies waste into **7 simple categories** (*Battery, Cardboard, Clothes, Glass, Metal, Paper, Plastic*). It also connects with an easy web app where users earn points and see how much CO2 gas they save by recycling.

2. Data & Preprocessing Pipeline

- **Dataset Split:** Split data into **80% for training** and **20% for testing/validation** using a fixed seed (123).
- **Image Preprocessing:** Resized all photos to 224x224 pixels and normalized pixel values to [-1, 1].
- **Data Augmentation:** Applied horizontal flips, 20% rotation, and 20% zoom so the model recognizes photos from any angle.
- **Fast Loading:** Used tf.data.AUTOTUNE and caching to speed up data loading during training.

3. Three Key Technical Decisions

Decision 1: MobileNetV2 Pre-trained Model
Instead of training a model from scratch, we used MobileNetV2. It gives high accuracy while keeping the file very small (around 11 MB), making predictions super fast on web browsers.

Decision 2: Live Data Augmentation in Keras
Added rotation, flip, and zoom layers directly inside the model. This creates new variations of images on-the-fly during training without filling up disk storage.

Decision 3: Interactive Streamlit Web App & Points System
Built a user-friendly app with Streamlit where users log in, upload photos, get instant predictions, collect eco-points, and track their carbon savings.